Exercise 3.4

Required

Textbook p. 85

Textbook sections: 3.1.1

1. Wide-sense interpretation.

Independence here means independence of the two processes, including their finite vectors, not merely independence of $X(t)$ and $Y(t)$ at one time. If their constant means are m_X, m_Y, independence gives

$$\boxed{m_Z = m_X m_Y}, \qquad \begin{aligned} R_Z(t+\tau, t) &= \mathbb{E}[X(t+\tau)X(t)Y(t+\tau)Y(t)] \\ &= R_X(\tau)R_Y(\tau). \end{aligned}$$

The second moments are finite because $\mathbb{E}[Z^2(t)] = \mathbb{E}[X^2(t)]\mathbb{E}[Y^2(t)]$. Thus Z has a constant mean and a lag-only correlation, so it is WSS.

2. Strict-sense interpretation.

For arbitrary $t_1, \ldots, t_k$, write $\boldsymbol{X} = (X(t_i))$ and $\boldsymbol{Y} = (Y(t_i))$. The joint law is the product of their laws. A common time shift leaves each law invariant and preserves this product. Applying the deterministic map

$$(\boldsymbol{x}, \boldsymbol{y}) \longmapsto (x_1 y_1, \ldots, x_k y_k)$$

therefore leaves the law of $(Z(t_1), \ldots, Z(t_k))$ invariant. Hence independent strictly stationary processes also have a strictly stationary product. No density assumption is needed.

Exercise 3.5

Required

Textbook p. 85

Textbook sections: 3.1.1 / 3.3.1

1. Use a harmonic identity.

Let $a = \omega_0 t + \Theta$. Then $Y(t) = 100 \sin^2 a = 50 - 50\cos(2a)$. A uniform phase gives $\mathbb{E}[\cos(2a)] = 0$, hence $\boxed{m_Y = 50}$.

2. Multiply the two samples before averaging.

For $a_1 = \omega_0(t+\tau) + \Theta$ and $a_0 = \omega_0 t + \Theta$,

$$\mathbb{E}[\cos(2a_1)\cos(2a_0)] = \tfrac{1}{2}\cos(2\omega_0\tau),$$

because the sum-angle term averages to zero. Consequently,

$$\boxed{R_Y(\tau) = 2500 + 1250\cos(2\omega_0\tau)}.$$

The constant mean and lag-only correlation prove WSS. Also $\mathbb{E}[Y^2] = 3750$, $\mathrm{Var}(Y) = 1250$ and $C_Y(\tau) = 1250\cos(2\omega_0\tau)$. The same formulas hold for $\omega_0 = 0$, when every realization of Y is constant in time.

Exercise 3.9

Required

Textbook p. 85

Textbook sections: 3.1.2 / 3.3.1

1. Remove cross terms using independence and zero means.

For all s, t, $R_{XY}(s,t) = \mathbb{E}[X(s)]\mathbb{E}[Y(t)] = 0$, and likewise $R_{YX} = 0$.

2. Expand each product.

$$\begin{aligned} R_Z(\tau) &= R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau), \\ R_W(\tau) &= R_X(\tau) + R_Y(\tau) - R_{XY}(\tau) - R_{YX}(\tau), \\ R_{ZW}(\tau) &= R_X(\tau) - R_Y(\tau) - R_{XY}(\tau) + R_{YX}(\tau). \end{aligned}$$

Using $R_{ZW}(\tau) = \mathbb{E}[Z(t+\tau)W(t)]$ therefore yields

$$\boxed{R_Z(\tau) = R_W(\tau) = \mathrm{e}^{-|\tau|} + \cos(2\pi\tau)},$$

$$\boxed{R_{ZW}(\tau) = \mathrm{e}^{-|\tau|} - \cos(2\pi\tau)}.$$

Here $R_Z(0) = R_W(0) = 2$ and $R_{ZW}(0) = 0$. Zero same-time covariance does not by itself imply independence of Z and W.

Exercise 3.14

Required

Textbook p. 86

Textbook sections: 3.3.1 / 3.4.2

1. State the tests and transform convention.

For a real WSS signal, R is real and even, $R(0) = \sigma^2 + m^2$, and it is positive semidefinite. A nonnegative finite spectral measure is a sufficient validity test. We use

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau)\mathrm{e}^{-\mathrm{j}\omega\tau}\, d\tau, \qquad R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\mathrm{e}^{\mathrm{j}\omega\tau}\, d\omega.$$

2. The four rejected candidates.

1. Not valid: $5u(\tau)\mathrm{e}^{-3\tau}$ is not even, regardless of the convention for $u(0)$.
2. Not valid: $5\sin(5\tau)$ is odd and has $R(0) = 0 \neq 5$.
3. Not valid for the specified Y: its value at zero is $9 \neq 10$. It is a valid correlation for a zero-mean process of variance 9, so rejection is due to the given variance.
4. Not valid: $R(0) = -1$, which cannot be a mean-square value.

3. The four accepted candidates.

Write $\Lambda(x) = \max(1-|x|, 0)$; rectangle edge values do not affect the integrals. Each of the following spectra is a nonnegative finite measure:

$$\begin{aligned} S_{(5)}(\omega) &= \frac{5\pi}{3}\Lambda(\omega/6), \\ S_{(6)}(\omega) &= 12\pi\delta(\omega) + \frac{2\pi}{5}\mathbf{1}_{\{|\omega|<10\}}, \\ S_{(7)}(\omega) &= \frac{10}{1+\omega^2}, \\ S_{(8)}(\omega) &= 12\pi\delta(\omega) + 4\sqrt{\frac{\pi}{3}}\mathrm{e}^{-\omega^2/12}. \end{aligned}$$

They have $R_{(5)}(0) = R_{(7)}(0) = 5$ and $R_{(6)}(0) = R_{(8)}(0) = 10$. Thus $\boxed{\text{(5), (6), (7), (8) are valid.}}$ The constant 6 in (6) and (8) can come from a time-constant zero-mean random component of variance 6; it does not force the ensemble mean to be nonzero. Evenness and the bound $|R(\tau)| \le R(0)$ alone would not establish validity.

Exercise 3.16

Required

Textbook p. 86

Textbook sections: 3.1.1 / 3.3.1

1. Determine the mean of X rather than assuming it.

The supplied R_X is absolutely integrable and has no zero-frequency spectral atom. One direct justification uses $\overline{X}_T = T^{-1}\int_0^T X(t)\,dt$. Since $|R_X(\tau)| \le 2\mathrm{e}^{-|\tau|}$,

$$m_X^2 = |\mathbb{E}\overline{X}_T|^2 \le \mathbb{E}[\overline{X}_T^2] \le \frac{1}{T^2}\int_0^T\int_0^T 2\mathrm{e}^{-|t-s|}\,ds\,dt \le \frac{4}{T}.$$

Letting $T \to \infty$ gives $m_X = 0$. The continuous correlation ensures that this mean-square averaging argument is legitimate. We need not determine m_Y.

2. Keep the same random amplitude at both times.

$\mathbb{E}[A^2] = \mathrm{Var}(A) + (\mathbb{E}A)^2 = 9 + 4 = 13$. Mutual independence gives

$$\boxed{m_Z = \mathbb{E}[A]m_X m_Y = 0},$$

$$\begin{aligned} R_Z(\tau) &= \mathbb{E}[A^2]R_X(\tau)R_Y(\tau) \\ &= \boxed{26\mathrm{e}^{-|\tau|}\cos(\omega_0\tau)\big(9 + \mathrm{e}^{-3\tau^2}\big)}. \end{aligned}$$

Finally, $\boxed{\mathrm{Var}[Z(t)] = R_Z(0) - m_Z^2 = 260}$. Replacing $\mathbb{E}[A^2]$ by $(\mathbb{E}A)^2$ would incorrectly treat the two occurrences of A as independent draws.

Exercise 3.21

Required

Textbook p. 86–87

Textbook sections: 3.4.2

1. Apply the real-signal PSD criteria.

An ordinary finite-power real WSS signal has a real, even, nonnegative spectral measure of finite total mass. A density must therefore be integrable; impulse weights must be nonnegative.

1. Valid. The removable value at $\omega = 0$ is 1. It is real, even, nonnegative and integrable, decaying as $O(\omega^{-2})$.
2. Valid. The denominator $\omega^6 + 3\omega^2 + 3$ is strictly positive for real ω. The function is even, nonnegative, regular at zero and $O(\omega^{-4})$ at infinity.
3. Invalid. For $0 < |\omega| < 1$ the ordinary part is negative; there are poles at $\omega = \pm 1$ and a negative impulse at zero. These already violate nonnegativity.
4. Invalid. For nonzero real ω, the denominator generally has a nonzero imaginary part; the function is not a real PSD.
5. Valid. It equals $|\omega|/(1+\omega^2)^2$, an even nonnegative integrable function. Its integral is 1, so its mean-square value would be $1/(2\pi)$.
6. Invalid for a real signal. It is nonnegative and integrable but is not even: its center is $\omega = 1$. It could be a PSD for a complex signal, which is outside the stated question.

Thus $\boxed{(1),\ (2),\ (5) \text{ are valid.}}$

Exercise 3.25

Required

Textbook p. 87

Textbook sections: 3.4.2

1. Transform the prescribed cross-correlation.

Using $R_{XY}(\tau) = \mathbb{E}[X(t+\tau)Y(t)]$ and the forward kernel $\mathrm{e}^{-\mathrm{j}\omega\tau}$,

$$S_{XY}(\omega) = 9\int_0^\infty \mathrm{e}^{-(3+\mathrm{j}\omega)\tau}\,d\tau = \boxed{\frac{9}{3+\mathrm{j}\omega}}.$$

The single endpoint value at $\tau = 0$ does not affect this Fourier integral.

2. Reverse the process order.

For real jointly WSS processes, $R_{YX}(\tau) = R_{XY}(-\tau)$, hence

$$\boxed{S_{YX}(\omega) = S_{XY}^*(\omega) = \frac{9}{3-\mathrm{j}\omega}}.$$

A cross-spectrum may be complex and need not be nonnegative. Its real part here is $27/(9+\omega^2)$ and the imaginary part of S_{XY} is $-9\omega/(9+\omega^2)$. These are the requested formal transforms; individual auto-spectra were not supplied for a joint spectral-realizability test.

Exercise 3.1

Optional

Textbook p. 85

Textbook sections: 3.1.1

1. Distinct observation times.

For each t, $U(t) \sim N(0,2)$, so

$$\boxed{f(u;t) = \frac{1}{2\sqrt{\pi}} \mathrm{e}^{-u^2/4}}.$$

If $t_1 \neq t_2$, independence gives

$$\boxed{f(u_1,u_2;t_1,t_2) = \frac{1}{4\pi} \mathrm{e}^{-(u_1^2+u_2^2)/4}}.$$

For pairwise distinct $t_1,\dots,t_k$, the stated independence of samples gives

$$\boxed{f(u_1,\dots,u_k;t_1,\dots,t_k) = (4\pi)^{-k/2} \exp\left(-\frac{1}{4}\sum_{i=1}^{k} u_i^2\right)}.$$

2. Repeated times and stationarity.

Repeated times refer to the same random variable, not independent draws. For example, if $t_1 = t_2$ the law is supported on $u_1 = u_2$, with generalized density $f(u_1;t_1)\delta(u_2-u_1)$; an ordinary two-dimensional density does not exist. For general repeated times, take a product density over distinct time groups and impose equality inside each group. All finite-dimensional laws depend only on which times coincide. A common time shift preserves those equalities, proving strict stationarity. Also

$$m_U = 0, \qquad R_U(\tau) = \begin{cases} 2, & \tau = 0, \\ 0, & \tau \neq 0, \end{cases}$$

so the process is WSS. This continuum-indexed independent-sample model is not mean-square continuous. Do not replace this finite-variance correlation by a Dirac impulse: ideal continuous-time white noise is a different generalized-process model.

Exercise 3.7

Optional

Textbook p. 85

Textbook sections: 3.2.2

1. Shift both correlation arguments by the same number of cycles.

Wide-sense cyclostationarity gives $m(t+100k) = m(t)$ and

$$R(t_1+100k, t_2+100k) = R(t_1,t_2), \qquad k \in \mathbb{Z}.$$

It does not in general permit arbitrary independent shifts of the two arguments. Linearity of expectation needs no independence assumption.

2. Substitute the supplied mean and correlation.

$$\begin{aligned}
\mathbb{E}[5Y(110)] &= 5m(10) = \boxed{100},\\
\mathbb{E}[10Y(310)+50] &= 10m(10)+50 = \boxed{250},\\
\mathbb{E}[Y(105)Y(101)] &= R(5,1) = \boxed{10},\\
\mathbb{E}[30Y(205)Y(201)+200] &= 30R(5,1)+200 = \boxed{500},\\
\mathbb{E}[10Y(305)Y(301)+6Y(210)+80] &= 10R(5,1)+6m(10)+80 = \boxed{300}.
\end{aligned}$$

Exercise 3.19

Optional

Textbook p. 86

Textbook sections: 3.4.2

1. Rational spectrum: partial fractions.

Factor $\omega^4+3\omega^2+2=(\omega^2+1)(\omega^2+2)$ and write

$$S_X(\omega) = -\frac{1}{\omega^2+1}+\frac{2}{\omega^2+2}.$$

The pair $\mathcal{F}\{\mathrm{e}^{-a|\tau|}\} = 2a/(a^2+\omega^2)$ gives

$$\boxed{R_X(\tau) = -\frac{1}{2}\mathrm{e}^{-|\tau|}+\frac{1}{\sqrt{2}}\mathrm{e}^{-\sqrt{2}|\tau|}}, \qquad \boxed{\mathbb{E}[X^2] = \frac{\sqrt{2}-1}{2}}.$$

A negative tail of R_X is allowed; its Fourier transform is still nonnegative.

2. Impulse plus triangular spectrum.

Write the spectrum as $8\delta(\omega)+20\Lambda(\omega/10)$. An impulse at zero gives the constant $8/(2\pi)=4/\pi$. The inverse transform of the triangle is a squared sinc, giving

$$\boxed{R_X(\tau) = \frac{4}{\pi}+\frac{100}{\pi}\left[\frac{\sin(5\tau)}{5\tau}\right]^2}, \qquad \boxed{\mathbb{E}[X^2] = \frac{104}{\pi}}.$$

The continuous triangle has area $20\times 10=200$; with impulse mass 8, the total power is $(200+8)/(2\pi)$.

3. The nondecaying spectrum requires a generalized interpretation.

If $p>0$, $p\cos^4(\omega\tau_0)$ has an infinite integral (also when $\tau_0=0$). Thus it is not the spectrum of an ordinary finite-variance WSS signal and there is no finite mean-square value. Using

$$\cos^4 x = \frac{3}{8}+\frac{1}{2}\cos 2x+\frac{1}{8}\cos 4x,$$

its inverse transform as a tempered distribution is

$$\boxed{\begin{aligned}R_X(\tau) = p\big\{&\tfrac{3}{8}\delta(\tau)+\tfrac{1}{4}[\delta(\tau-2\tau_0)+\delta(\tau+2\tau_0)]\\ &+\tfrac{1}{16}[\delta(\tau-4\tau_0)+\delta(\tau+4\tau_0)]\big\}.\end{aligned}}$$

Distributional $R_X(0)$ is not a finite number; writing $\delta(0)$ as a numeric answer is invalid. If $p=0$, the zero signal has zero power. If $p<0$, the expression is negative on a set of positive measure and cannot be a PSD. The original question did not restrict p, so these cases must be separated.